

6(a)	(number density =) $2.3 \times 10^{23} / (1.5 \times 10^{-6} \times 1.8) = 8.5 \times 10^{28} \text{ (m}^{-3}\text{)}$	A1
6(b)	$I = Q/t$ or $I = 172 / 2.5 \times 60$ or $I = 1.1(5)$	C1
	$I = nAvq$ $172 / (2.5 \times 60) = 1.5 \times 10^{-6} \times 8.5 \times 10^{28} \times v \times 1.6 \times 10^{-19}$	C1
	$v = 5.6 \times 10^{-5} \text{ m s}^{-1}$	A1